# CEUE203 – Object Oriented Programming

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## Practical – 5 (Part A )

### Aim

To understand abstract classes, inheritance, method overriding, and runtime polymorphism using a simple Shape example in Java.

### Program Code

```
abstract
class Shape {     abstract
double area();
}

class Circle extends Shape {
double r;     Circle(double
r) {        this.r = r;
   }
   double area() {
return 3.14 * r * r;
   }
}

class Rectangle extends Shape {
double l, b;
   Rectangle(double l, double b) {
      this.l = l;
this.b = b;
   }
   double area() {
return l * b;
   }
}

public class Practical5 {
   public static void main(String[] args) {
Shape[] s = {           new Circle(2),
new Rectangle(4, 5)
      };

      for (Shape x : s) {
         System.out.println("Area = " + x.area());
```

```
    }
  }
}
```

### Output

Expected Output:
Area = 12.56
Area = 20.0

### Learning Outcome

I learned how to create an abstract class, inherit it using subclasses, override methods, and use polymorphism to call different area() methods through a Shape array.